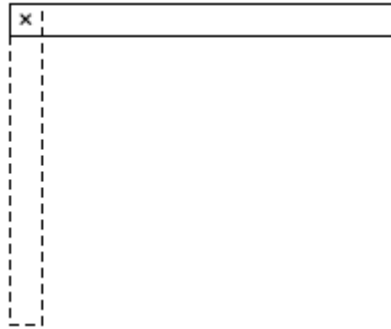


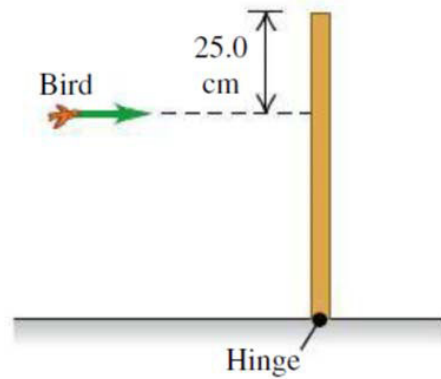
1. A computer disk drive is turned on starting from rest and has constant angular acceleration. If it took 0.610 s for the drive to make its second complete revolution,
 - (i) how long did it take to make the first complete revolution (1.47 s), and
 - (ii) what is its angular acceleration, in rad/s^2 ? (5.79 rad/s^2)

2. Calculate the position of the new axis of rotation for a uniform balsa-wood solid sphere where its moment of inertia with respect to the new axis has the same value, when compared to the moment of inertia of a thin-walled hollow lead sphere with respect to its axis of rotation along its diameter. Assume that both solid object and hollow object have the same mass and same radius. $\left(\sqrt{\frac{4}{15}}R \text{ from diameter}\right)$

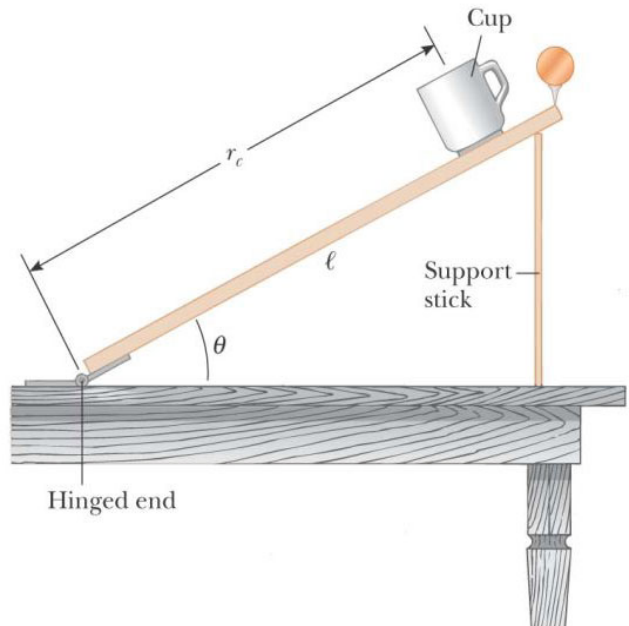
3. A metre stick with a mass of 0.170 kg is pivoted about one end so it can rotate without friction about a horizontal axis. The metre stick is held in a horizontal position and released. As it swings through the vertical, calculate
 - (i) the angular speed of the stick; (5.42 rad/s)
 - (ii) the linear speed of the end of the stick opposite the axis. Compare your answer to the speed of a particle that has fallen 1.00 m starting from rest. (5.42 m/s, 4.43 m/s)



4. A 460.0-g bird is flying horizontally at 2.30 m/s to a stationary vertical bar, and it hits the bar 25.0 cm below the top as shown in the figure. The bar is uniform, and has a mass of 2.50 kg and length of 0.78 m. It is hinged at its base. The collision stuns the bird so that it subsequently drops vertically to the ground. What is the angular velocity of the bar (i) just after it is hit by the bird, and, (ii) just as it reaches the ground. (1.11 rad/s , 6.24 rad/s)

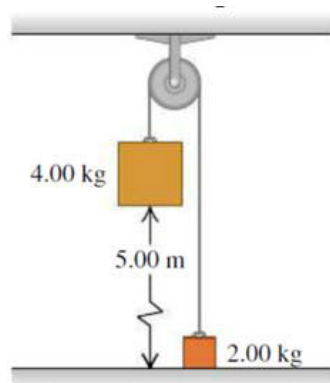


5. The figure consists of a ball resting at one end of a uniform board of length ℓ , hinged at the other end, and elevated at an angle θ . A light cup is attached to the board at r_c so that it will catch the ball when the support stick is suddenly removed.



- (i) Show that the ball will lag behind the falling board when θ is less than 35.3° .
- (ii) If the board is 1.00 m long and is supported at this limiting angle, show that the cup must be 18.4 cm from the moving end.

6. The frictionless pulley in the figure has radius 0.160 m and moment of inertia 0.380 kg m^2 . The flexible rope has negligible mass and does not slip on the pulley rim.

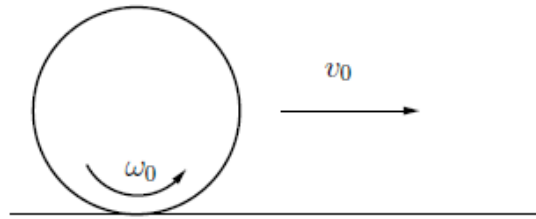


Figure

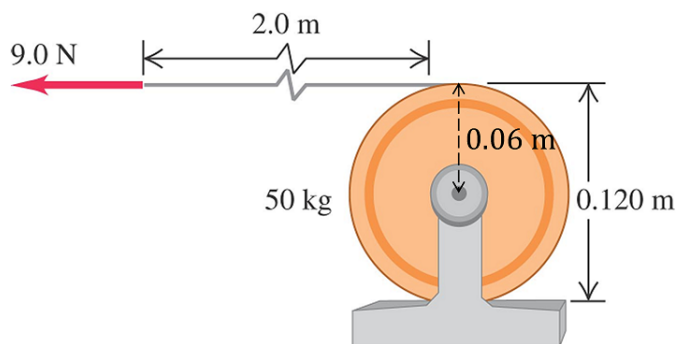
- (i) Determine the linear accelerations of the 2.00 kg and 4.00 kg blocks, the angular acceleration of the pulley, and the tension in each side of the rope. (35.44 N, 21.48 N, 5.88 m/s²)

- (ii) Determine the speed of the 4.00 kg block just before it strikes the floor. (3.07 m/s)
 (iii) The pulley is a solid uniform disk and is supported by sort of a light “hook” connected to the ceiling. What force does the ceiling exert on the “hook”? (347.86 N)

7. A billiard ball is sliding on a table top with a horizontal velocity v_0 to the right and an anti-clockwise angular speed ω_0 about its center of mass as shown in figure when the time $t = 0$. The ball is a uniform solid sphere of radius R and mass M . The coefficient of kinetic friction between the ball and the table is μ_k .

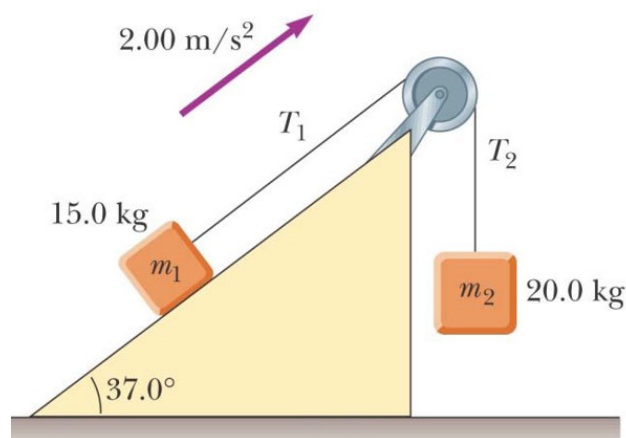


- (i) Show that if $v_0 < \frac{2}{5} R\omega_0$, the ball will be still rolling anti-clockwise when its center of mass first comes to an instantaneous stop in its linear displacement.
- (ii) Describe qualitatively the subsequent motion of the ball in (i) if $v_0 < \frac{2}{5} R\omega_0$.
- (iii) Determine the time taken by the ball from $t = 0$ before its rolling without slipping starts if $v_0 < \frac{2}{5} R\omega_0$. $\left(\frac{2}{7\mu_k g} (v_0 + R\omega_0) \right)$
8. Steel train rails are laid in 11.0-m-long segments placed end to end. The rails are laid on a winter day when the temperature is -2.5°C . (i) How much space must be left between adjacent rails if they are just to touch on a summer day when their temperature is 35.0°C ? (ii) If the rails are originally laid in contact, what is the stress in them on a summer day when their temperature is 35.0°C ? Is this stress compressive or tensile?
 $\left(\alpha_{\text{Steel}} = 1.2 \times 10^{-5} (\text{C})^{-1}, Y_{\text{Steel}} = 2.0 \times 10^{11} \text{ N/m}^2 \right)$ (4.95 mm, $-9 \times 10^7 \text{ N/m}^2$)
9. In a circus performance a large 5.0-kg hoop of radius 3.0 m rolls without slipping. If the hoop is given an angular speed of 3.0 rad/s while rolling on the horizontal ground and is then allowed to roll up a ramp inclined at 20° with the horizontal, how far along the incline does the hoop roll? (24.17 m)
10. We wrap a light and nonstretching cable around a solid cylinder of mass 50.0 kg and diameter 0.120 m, which rotates in frictionless bearings about a stationary horizontal axis. Subsequently we pull the free end of the cable with a constant 9.00 N force for a distance of 2.00 m; it turns the cylinder as it unwinds without slipping. The cylinder is initially at rest.



- (i) What is the cable's acceleration a ? (0.36 m/s^2)
- (ii) Find the final speed v of the cable and the final angular speed ω of the solid cylinder. (1.2 m/s , 20 rad/s)

11. Two blocks, as shown in the Figure, are connected by a string of negligible mass passing over a pulley of radius 0.250 m and moment of inertia I . The block on the rough inclined surface is moving up with a constant acceleration of 2.00 m/s^2 . The coefficients of friction for the inclined surface are $\mu_s = 0.1$ and $\mu_k = 0.06$, while the vertical surface is smooth.

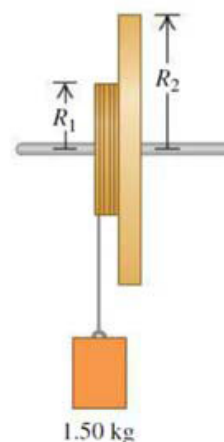


Assume that the string does not slip relative to the pulley. Determine T_1 and T_2 , which are the tensions in the two parts of the string, and the moment of inertia (I) of the pulley.

(156 N , 125.51 N , 0.95 kg m^2)

12. Two metal disks, one with radius $R_1 = 2.55 \text{ cm}$ and mass $M_1 = 0.850 \text{ kg}$ and the other with radius $R_2 = 5.02 \text{ cm}$ and mass $M_2 = 1.58 \text{ kg}$, are welded together and mounted on a frictionless axis through their common centre, as shown in the figure.

- (i) What is the total moment of inertia of the two disks?
($2.27 \times 10^{-3} \text{ kg m}^2$)
- (ii) A light string is wrapped around the edge of the smaller disk, and a 1.50 kg block is suspended from the free end of the string. If the block is released from rest at a distance of 2.03 m above the floor, what is its speed just before it strikes the floor?
(3.46 m/s)
- (iii) Repeat part (ii), this time with the string wrapped around the edge of the larger disk. In which case is the final speed of the block greater? Explain. (4.99 m/s)



13. A brass rod is 185 cm long and 1.60 cm in diameter. What force must be applied to each end of the rod to prevent it from contracting when it is cooled from 120.0°C to 10.0 °C?
 $\left(\alpha_{\text{Brass}} = 2.0 \times 10^{-5} (\text{°C})^{-1}, Y_{\text{Brass}} = 0.9 \times 10^{11} \text{ N/m}^2 \right) \quad (3.98 \times 10^4 \text{ N})$

- End -

Formula List

Logarithm:

$$\log a + \log b = \log(a \times b)$$

$$\log a - \log b = \log\left(\frac{a}{b}\right)$$

$$(\log x) > 0, \text{ if } x > 1.$$

$$(\log 1) = 0.$$

$$(\log x) < 0, \text{ if } 0 < x < 1.$$

$$(\log x) \text{ is undefined if } x < 0.$$

Trigonometry:

$$\sin = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin}{\cos}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Differentiation:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \times \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(f(x) \times g(x)) = f(x) \times \left(\frac{d}{dx}g(x)\right) + \left(\frac{d}{dx}f(x)\right) \times g(x)$$

Integration:

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

Vector:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

$$\vec{A} \bullet \vec{B} = |\vec{A}| \times |\vec{B}| \times \cos \theta$$

Linear kinematic equations:

$$v_f = v_i + at$$

$$\Delta x = v_i t + \frac{1}{2} at^2$$

$$\Delta x = \frac{(v_i + v_f)}{2} \times t$$

$$v_f^2 = v_i^2 + 2a\Delta x$$

Simple harmonic motion:

$$x(t) = r_0 \cos(\theta_0 + \omega_0 t)$$

$$v_x(t) = -r_0 \omega_0 \sin(\theta_0 + \omega_0 t)$$

$$a_x(t) = -r_0 \omega_0^2 \cos(\theta_0 + \omega_0 t) \\ = -\omega_0^2 x(t)$$

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$v_{\max} = \sqrt{\frac{k}{m}} A$$

$$a_{\max} = \frac{k}{m} A$$

Retarding force:

$$R = -bv$$

Projectile motion:

$$y = -\frac{g}{2v_{0x}^2}(x - x_0)^2 + \frac{v_{0y}}{v_{0x}}(x - x_0) + y_0, \text{ or}$$

$$y = x \tan \theta - \frac{gx^2}{2v_0^2}(1 + \tan^2 \theta)$$

Uniform circular motion:

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T}$$

$$v = r\omega$$

$$a_c = r\omega^2 = \frac{v^2}{r}$$

Non-uniform circular motion:

$$a_t = \frac{dv}{dt}$$

$$a = \sqrt{a_c^2 + a_t^2}$$

Relative velocity:

$$\vec{V}_{P/A} = \vec{V}_{P/B} + \vec{V}_{B/A}$$

Newton's second law:

$$\sum \vec{F} = m\vec{a}$$

Newton's third law:

$$\vec{F}_{12} = -\vec{F}_{21}, \text{ and the two forces} \\ \text{act on the opposite body.}$$

Frictional force:

$$f_s \leq \mu_s n, \text{ and impending motion} \\ \text{occurs when } f_s = \mu_s n.$$

$$f_k = \mu_k n$$

Translational kinetic energy:

$$KE_{\text{translational}} = \frac{1}{2}mv^2$$

Gravitational potential energy:

$$\Delta PE = mg(y_2 - y_1) \\ = mgh$$

Spring

$$F_{\text{spring}} = -kx$$

$$\text{Elastic } PE_{\text{spring}} = \frac{1}{2}kx^2$$

Work-energy theorem:

$$W = \Delta KE = KE_f - KE_i$$

Work done (constant force):

$$W = \vec{F} \bullet \Delta \vec{r} = F \Delta r \cos \theta$$

Work done (varying force):

$$W = \int_{x_i}^{x_f} \vec{F} \bullet d\vec{r}$$

Power:

$$P = \frac{dW}{dt} = F \times v$$

Linear momentum:

(also valid for y and z)

$$p_x = mv_x$$

$$F_x = \frac{dp_x}{dt}$$

$$F_{\text{average, x}} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} F_x(t) dt$$

Conservation of linear momentum:

$$\vec{p}_{1,i} + \vec{p}_{2,i} = \vec{p}_{1,f} + \vec{p}_{2,f}$$

Elastic collision:

$$v_{A1} - v_{B1} = -(v_{A2} - v_{B2})$$

$$v_{A2} = \left(\frac{m_A - m_B}{m_A + m_B} \right) v_{A1} + \left(\frac{2m_B}{m_A + m_B} \right) v_{B1}$$

$$v_{B2} = \left(\frac{m_B - m_A}{m_A + m_B} \right) v_{B1} + \left(\frac{2m_A}{m_A + m_B} \right) v_{A1}$$

Center of mass

(also valid for y and z)

$$x_{\text{cm}} = \frac{1}{M} \sum m x, \text{ where } M = \sum m, \text{ or}$$

$$= \frac{1}{M} \int x \, dm$$

$$v_{\text{cm},x} = \frac{1}{M} \sum m v_x$$

$$a_{\text{cm},x} = \frac{1}{M} \sum m a_x$$

Simple Harmonic Motion:

$$a_x = -\omega^2 x$$

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$v_{\text{max}} = \sqrt{\frac{k}{m}} A$$

$$a_{\text{max}} = \frac{k}{m} A$$

Retarding force:

$$F = -b v$$

Rotation

$$\theta = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

$$v_{\text{tangential}} = r\omega$$

$$a_{\text{centripetal}} = r\omega^2 = \frac{v^2}{r}$$

$$a_{\text{tangential}} = r\alpha$$

Rotational kinematic equations:

$$\omega_f = \omega_i + \alpha t$$

$$\Delta\theta = \omega_i t + \frac{1}{2} \alpha t^2$$

$$\Delta\theta = \frac{(\omega_i + \omega_f)}{2} \times t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

Moment of inertia:

$$I = \sum m r^2$$

$$I = \int r^2 \, dm$$

Parallel axis theorem

$$I = I_{\text{cm}} + M d^2$$

Rotational kinetic energy:

$$KE_{\text{rotational}} = \frac{1}{2} I \omega^2$$

Moment of inertia:

$$I = \sum m_i r_i^2$$

$$I = \int r^2 \, dm$$

Moment of inertia of hoop or thin cylindrical shell:

$$I_{\text{CM}} = MR^2$$

Moment of inertia of hollow cylinder:

$$I_{\text{CM}} = \frac{1}{2} M (R_1^2 + R_2^2)$$

Moment of inertia of solid cylinder:

$$I_{\text{CM}} = \frac{1}{2} MR^2$$

Moment of inertia of rectangular plate:

$$I_{\text{CM}} = \frac{1}{12} M (a^2 + b^2)$$

Moment of inertia of thin rod about its CM:

$$I_{CM} = \frac{1}{12} ML^2$$

Moment of inertia of thin rod about its end:

$$I = \frac{ML^2}{3}$$

Moment of inertia of solid sphere:

$$I_{CM} = \frac{2}{5} MR^2$$

Moment of inertia of spherical shell:

$$I_{CM} = \frac{2}{3} MR^2$$

Torque and Moment of force:

$$\tau = F \times r \times \sin \theta$$

$$\sum \tau = I\alpha$$

Work due to rotation:

$$\Delta W = \tau \times \Delta \theta$$

$$W = \int_{\theta_1}^{\theta_2} \tau d\theta$$

Power due to rotation:

$$P = \tau \times \omega$$

Pure rolling:

$$v_{cm} = r\omega$$

$$a_{cm} = r\alpha$$

$$KE_{\text{total}} = \frac{1}{2} Mv_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

Conservation of Angular momentum:

$$\underbrace{\sum I\omega}_{\text{Before collision}} = \underbrace{\sum I\omega}_{\text{After collision}}$$

Avogadro's number:

$$N_A \approx 6.022 \times 10^{23} \text{ particles/mole}$$

Ideal gas constant:

$$R \approx 8.314 \text{ J mole}^{-1} \text{ K}^{-1}$$

Boltzmann constant:

$$k = \frac{R}{N_A} \approx 1.381 \times 10^{-23} \text{ J/K}$$

Ideal gas equation:

$$pV = nRT$$

Speed of a molecule

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

Mole

$$n = \frac{N}{N_A}$$

$$n = \frac{m}{M}$$

Absolute temperature

$$T = T_{\text{Celsius}} + 273.15$$

Specific heat c and molar heat C

$$Q = Mc(T_2 - T_1)$$

$$= nC(T_2 - T_1)$$

Latent heat L

$$Q = L \times M$$

First law of thermodynamics

$$\Delta U = Q - W$$

Internal energy for N molecules

$$\text{monatomic: } U = N \times \frac{3}{2} kT$$

$$\text{diatomic: } U = N \times \frac{5}{2} kT$$

Relationship between C_p and C_v for an ideal Gas

$$C_p - C_v = R$$

Isochoric process

$$\text{monatomic: } C_V = \frac{3}{2}R$$

$$\text{diatomic: } C_V = \frac{5}{2}R$$

C_p is undefined for isochoric process.

Isobaric process

$$\text{monatomic: } C_p = \frac{5}{2}R$$

$$\text{diatomic: } C_p = \frac{7}{2}R$$

C_V is undefined for isobaric process.

Isothermal process

$$W = nRT \ln \left(\frac{V_f}{V_i} \right)$$

Adiabatic process

$$\gamma = \frac{C_p}{C_V}$$

$$\text{monatomic: } \gamma = \frac{5}{3}$$

$$\text{diatomic: } \gamma = \frac{7}{5}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$P_1^{1-\gamma} T_1^\gamma = P_2^{1-\gamma} T_2^\gamma$$

Thermal efficiency of a heat engine

$$e = 1 - \frac{|Q_C|}{|Q_H|}$$

Coefficient of performance of a heat pump for cooling

$$\text{COP} = \frac{|Q_C|}{|Q_H| - |Q_C|}$$

Entropy

$$\Delta S = \int_{\text{initial state}}^{\text{final state}} \frac{dQ}{T} \geq 0$$

Coulomb's Law:

$$F = \frac{k q_1 q_2}{r^2}$$

where

$$k = 8.9875 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Electric field:

$$E = \frac{F}{q_0}$$

Electric field for point charge:

$$E = \frac{kq}{r^2}$$

Electric field at a distance along central axis of charged ring:

$$E = \frac{kQx}{(x^2 + a^2)^{3/2}}$$

Electric potential:

$$V = \frac{U}{q_0}$$

Electric potential in uniform field:

$$\Delta V = -Ed$$

Electric potential for point charge:

$$V = \frac{kq}{r}$$

Electric potential at a distance along central axis of charged ring:

$$V = \frac{kQ}{\sqrt{x^2 + a^2}}$$

Electric potential energy (for multiple point charges):

$$U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

Electric current:

$$I = \frac{dQ}{dt}$$
$$I_{av} = nqv_d A$$

Ohm's Law:

$$I = \frac{\Delta V}{R}$$

Resistance of ohmic conductor:

$$R = \rho \frac{L}{A}$$

Resistivity:

$$\rho = \rho_0 [1 + \alpha(T - T_0)]$$

Electric power:

$$P = I\Delta V = I^2 R = \frac{\Delta V^2}{R}$$

Electric energy:

$$U = P\Delta t$$

Terminal voltage:

$$\Delta V = \mathcal{E} - Ir$$

Resistors in series:

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

Resistors in parallel:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Kirchhoff's Junction Rule:

$$\sum I_{in} = \sum I_{out}$$

Kirchhoff's Loop Rule:

$$\sum \Delta V = 0$$

Magnetic force on a moving charge:

$$F_B = qvB \sin \theta$$

Magnetic force on a current carrying conductor:

$$F = IlB \sin \theta$$

Radius of moving charged particle in uniform magnetic field:

$$r = \frac{mv}{qB}$$

Angular speed:

$$\omega = \frac{qB}{m}$$

Period:

$$T = \frac{2\pi m}{qB}$$

Velocity selector:

$$v = \frac{E}{B}$$

Magnetic field due to long straight conductor:

$$B = \frac{\mu_0 i}{2\pi a} \text{ where}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

Magnetic field at center of circular wire loop:

$$B = \frac{\mu_0 i}{2R}$$

Magnetic field at a distance along central axis of current-carrying ring:

$$B = \frac{\mu_0 i R^2}{2(x^2 + R^2)^{3/2}}$$

Magnetic force between parallel conductors:

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi a}$$

Magnetic field inside toroid:

$$B = \frac{\mu_0 NI}{2\pi r}$$

Magnetic field inside solenoid:

$$B = \mu_0 nI$$

Magnetic flux:

$$\Phi_B = BA \cos \theta$$

Faraday's Law:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

Motional emf:

$$\mathcal{E} = \frac{Blv}{R}$$

Power input:

$$P = F_{\text{app}} v = I l B v$$